NATIONAL MATHEMATICS CONTEST
PRIMARY-LEVEL PAPER 2.

NAMES: _____ DATE OF BIRTH: _____

SCHOOL/INSTITUTION: _____

CLASS: _____ DISTRICT: _____

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

Do not turn this page until you have been instructed to do so. This competition is conducted on the assumption that proper security is maintained. Uganda Mathematical Society reserves the right, should there be evidence of any breach of security or any suspicion, to reject scripts from a particular school/institution.

PLEASE, NOTE CAREFULLY:

1. **WRITE YOUR ANSWERS IN THE SPACES PROVIDED. MORE SPACE IS AVAILABLE AT THE END OF THIS PAPER.**
2. Date: Saturday, 23rd June, 2018.
3. Time: Exactly 2 hours and 15 Minutes (from 9 : 00am to 11 : 15am)
4. Type:
   (a) This paper consists of two sections and each section carries 50 marks.
   (b) Attempt as many questions as you can. The total marks scored from all the questions you attempt will be your final score.
   (c) No marks will be awarded for an answer if NO clear and logical layout of the working is shown.
5. **Indicate your names, class, school and district on all your answer sheets**.
6. All diagrams/figures in this contest are not to scale.
7. Immediately the paper is done, the answer script(s) must be forwarded to:

   David Ddumba Walakira,
   Contest Coordinator, Uganda Mathematical Society,
   c/o Department of Mathematics, Makerere University,
   P.O. Box 7062, Kampala.
   Telephone contact: $+256-414-540692$.

   Answer scripts should be received **latest, Monday** 25**th June,** 2018.
8. **ALL** *participants who qualified and sat Paper* 2 *are cordially invited to the certificate and/or prize giving ceremony on Saturday* 28*th July,* 2018 ***at 8:00am*** *at Makerere University, Kampala.*
9. National Mathematics Contest 2019 Paper 1 will be done on 4th April, 2019 at various centres.
10. **UGANDA MATHEMATICAL SOCIETY** wishes you success in this year's contest.

SECTION A : 5 marks each

Qn. 1. What number should be added to 8 to give result -8?

Qn. 2. On a certain calculator, 2×10^4 is represented as $2E4$. How would the product of $2E4$ and $4E2$ be represented?

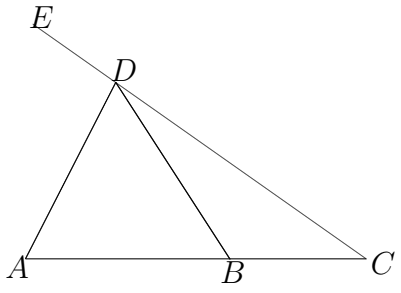
Qn. 3. Four points A, B, C, D are on a line segment in such a way that $AB : BC = 1 : 2$ and $BC : CD = 8 : 5$. Find $AB : BD$.

Qn. 4. Wilson is 15 and his sister Brenda is 7. In how many years will the sum of their ages be double what it is now?

Qn. 5. At Junior Academy, 30 girls and 20 boys wrote the 2017 mathematics contest. Certificates were awarded to 30% of the girls and 40% of boys. What percentage of all the participating students received certificates?

Qn. 6. A circle has an area of m cm^2 and a circumference of n cm. If $\frac{m}{n} = 20$, what is the diameter of the circle?

Qn. 7. In the figure, angle $DAB = 50^o$, angle $ADE = 80^o$ and angle $DBC = 125^o$. What is the value of angle BDC in the figure below?



Qn. 8. In a small town of 500 people, three children deliver all the newspapers except to 4 people. Abby delivers three times as many papers as Bob, and Connie delivers 13 more than Abby. How many papers does Connie deliver?

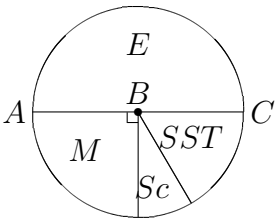
Qn. 9. Mike is a waiter at a restaurant. On Monday, he wakes up at 6.30 a.m, starts work at t a.m and finishes at t p.m. How long does Mike work on Monday?

Qn. 10. A swimming pool that measures 6m wide, 12m long and 4m deep is 75% full of water. What is the volume of water in the swimming pool?

SECTION B : 10 **marks each**

Qn. 11. Sarah walks at 4 km/hr and runs at 6 km/hr. She saves $3\frac{3}{4}$ minutes by running instead of walking from her home to school. What is the distance from her home to her school?

Qn. 12. ABC is the diameter of the circle centered at B. The circle illustrates the best subject (among E-English, M-Mathematics, Sc-Science and SST) for each of the 600 pupils of a certain primary school.



(a) If the proportion representing SST is equivalent to 60°, how many students have Science as their best subject?

(b) If 60% of those whose best subject is SST indeed passed SST with distinction 1, how many pupils passed SST with distinction 1?

Qn. 13. Kenneth and Lilian run around a track whose perimeter is 150m. It takes Kenneth 120 seconds to run 3 times around the track and it takes Lilian 160 seconds to run 5 times around the same track. Who of the two is faster and at what speed:

Qn. 14. A house maid charges Shs. 4,000 to babysit for the first one hour. For each additional hour, she charges 50% more than she charged for the previous hour. How much money in total would she earn for 5 hours of babysitting?

Qn. 15. Five students did a test that was marked out of 50. The scores of four of the students were 42, 43, 46 and 49. The score of the fifth student was X. The average score for the 5 students was the same as the median of the 5 students' scores. What are the possible values of X?

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SECTION A : 5 marks each

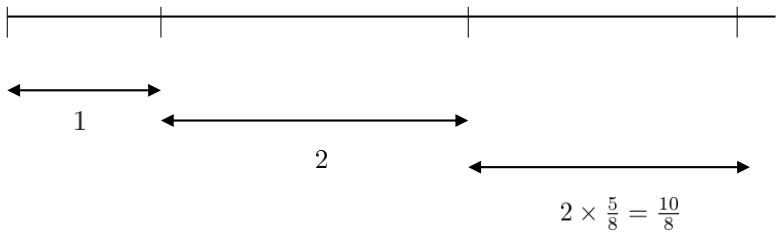
Qn. 1. $8 + x = -8 \quad \Rightarrow \quad x = -8 - 8 = -16$

Qn. 2. $(2E4)\,(4E2) = \left(2 \times 10^4\right)\left(4 \times 10^2\right) = 8 \times 10^6$ as $8E6$.

Qn. 3.

$$\begin{aligned} AB : BC &= 1 : 2 \\ BC : CD &= 8 : 5 \end{aligned}$$

to have



$$\begin{aligned} AB &: BD \\ 1 &: \left(2 + \frac{10}{8}\right) \\ 1 &: \frac{26}{8} \\ 8 &: 26 \\ 4 &: 13 \end{aligned}$$

Qn. 4. Let x be the number of years when the sum of Wilson and Brenda be double what it is now.

Currently: Sum of ages is $15 + 7 = 22$ years. In x year's time, Wilson will be $15 + x$ years old and Brenda will be $7 + x$ years;
The sum of their age then $(15 + x) + (7 + x)$ which is double their current age, that is

$$\begin{aligned} (15 + x) + (7 + x) &= 2(22) \\ 2x &= 44 - 22 = 22 \\ x &= \frac{22}{2} = 11 \text{ years.} \end{aligned}$$

Qn. 5.

$$\begin{aligned} \text{Number of girls that received certificates} &= \frac{30}{100} \times 30 = 9 \\ \text{Number of boys that received certificates} &= \frac{40}{100} \times 20 = 8 \\ \text{Total number of pupils that received certificates} &= 9 + 8 = 17 \\ \text{Total number of pupils} &= 30 + 20 = 50 \end{aligned}$$

The percentage that received certificates is $\frac{17}{50} \times 100 = 34\%$

Qn. 6.

$$\begin{aligned} \text{Area of a circle} \quad & m = \pi r^2 \\ \text{Circumference of a circle} \quad & n = 2\pi r \end{aligned}$$

From $\dfrac{m}{n} = 20 \quad \Rightarrow$

$$\begin{aligned} m &= 20n \\ \pi r^2 &= 20 \times 2\pi r \\ \frac{\pi r^2}{\pi r} &= \frac{40\pi r}{\pi r} \\ r &= 40 \end{aligned}$$

Therefore, the diameter is given by: Diameter $= 2r = 2 \times 40 = 80$cm.

Qn. 7. Let

$$\begin{aligned} \angle BDC &= x^\circ \\ \angle ADB &= y^\circ \\ \angle ABD &= z^\circ \end{aligned}$$

To have

$$\begin{aligned} z + 125^\circ = 108^\circ \quad &\Rightarrow \quad z = 180^\circ - 125^\circ = 55^\circ \\ y + z + 50^\circ = 180^\circ \quad &\Rightarrow \quad y = 180^\circ - 55^\circ - 50^\circ = 75^\circ \end{aligned}$$

$$\begin{aligned} x^\circ + 80^\circ + y^\circ &= 180^\circ \\ x^\circ + 80^\circ + 75^\circ &= 180^\circ \\ x^\circ + 155^\circ &= 180^\circ \\ x &= 180^\circ - 155^\circ = 25^\circ \end{aligned}$$

Thus angle $BDC = 25^\circ$.

Qn. 8. In a small town of 500 people, three children deliver all the newspapers except to 4 people. Abby delivers three times as many papers as Bob, and Connie delivers 13 more than Abby. How many papers does Connie deliver?

Let x be the papers delivered by Bob. Then $3x$ are the papers delivered by Abbey, and $(3x + 13)$ are the papers delivered by Connie. Therefore

$$x + 3x + (3x + 13) = 496 \quad \Rightarrow \quad 7x + 13 = 496 \quad \Rightarrow \quad 7x = 483 \quad \Rightarrow \quad x = 69$$

such that

$$\begin{aligned} \text{Abbey delevers} \quad & 3x = 3(69) = 207 \\ \text{Bob delevers} \quad & x = 69 \\ \text{Connie delevers} \quad & (3x + 13) = (207 + 13) = 220 \end{aligned}$$

Qn. 9. There are 12 hours between $t$am and $t$pm on the same day

Qn. 10. The volume of the swimming pool is given by

$$\begin{aligned} \text{Volume} &= l \times w \times h \\ &= 12 \times 6 \times 4 \\ &= 288m^3 \end{aligned}$$

75% full volume of water in pool is given by

$$\frac{75}{100} \times 288 = 216m^3$$

SECTION B : 10 **marks each**

Qn. 11. Let the distance between home be $d$km.

Speed:

$$\text{Walking speed} = 4km/hr \Rightarrow \text{Takes 1 hr to cover 4km, i.e,}$$

$$\begin{aligned} 4km &\text{ needs } 1hr \\ 1km &\text{ needs } \frac{1}{4}hrs \\ dkm &\text{ needs } d \times \frac{1}{4} = \frac{d}{4}hrs \end{aligned}$$

$$\text{Running speed} = 6km/hr \Rightarrow \text{Takes 1 hr to cover 6km, i.e,}$$

$$\begin{aligned} 6km &\text{ needs } 1hr \\ 1km &\text{ needs } \frac{1}{6}hrs \\ dkm &\text{ needs } d \times \frac{1}{6} = \frac{d}{6}hrs \end{aligned}$$

Therefore

$$\begin{aligned} \text{Time taken while walking} &= \frac{d}{4}hrs \\ \text{Time taken while running} &= \frac{d}{6}hrs \end{aligned}$$

Running saves $3\frac{3}{4} = 3.75$ minutes which is equal to $\frac{3}{60} + \frac{0.75}{60} = \frac{1}{20} + \frac{1}{80} = \frac{1}{16}$hrs. Therefore

$$\begin{aligned} \frac{d}{4} - \frac{d}{6} &= \frac{1}{16} \\ \frac{3d - 2d}{12} &= \frac{1}{16} \\ \frac{d}{12} &= \frac{1}{16} \\ 16d &= 12 \\ d &= \frac{12}{16} = \frac{3}{4} = 0.75km \end{aligned}$$

The distance between home and school is $\frac{3}{4}km = 0.75km$

Qn. 12. (a)

$$\begin{aligned} \angle M + \angle Sc + \angle SST &= 180^\circ \\ 90^\circ + \angle Sc + 60^\circ &= 180^\circ \\ \angle Sc &= 180^\circ - 90^\circ - 60^\circ \\ &= 30^\circ \end{aligned}$$

Proportion for science is $\frac{30^\circ}{360^\circ} = \frac{1}{12}$. Thus the number of students with Science as best subject is $\frac{1}{12} \times 600 = 50$ students.

(b) Proportion with SST as the best subject is $\frac{60^\circ}{360^\circ} = \frac{1}{6}$. Therefore, the number of students with SST as best subject is $\frac{1}{6} \times 600 = 100$ students. Those passing SST with distinction among the 100 are $\frac{60}{100} \times 100 = 60$ students.

Qn. 13. Kenneth runs 3 times of the 150m $\Rightarrow$ he covers $3 \times 150 = 450m$ this he covers in 120secs. Thus his speed $= \frac{450m}{120secs} = 3.75m/s$.

Lilian covers 5 times of the $150m \Rightarrow$ he covers $5 \times 150m = 750m$ in 160secs.

Thus his speed $= \frac{750m}{160secs} = 4.69m/s$

Thus Lilian is faster, at a speed of $4.69m/s$

Alternatively

Time per round used by Kennedy is $t_K = \frac{120}{3} = 40$ seconds per round

Time per round used by Lilian is $t_L = \frac{160}{5} = 32$ seconds per round

The time used by Lilian to cover a round is smaller than the time used by Kennedy, and thus, Lilian is the fastest at a speed of $s_L = \frac{D_L}{t_L} = \frac{150}{32} = 4.69$ meters per second

Qn. 14. Charging 50% more than previous hour $\Rightarrow$ If she charged $m$shs in previous hour, then the following next hour would cost $m + 0.5m = 1.5m \left(\frac{3}{2}m\right)$

$$\begin{aligned} 1^{st} \text{ hour} &= 4000 \\ 2^{nd} \text{ hour} &= 4000 \times \frac{3}{2} = 6000 \\ 3^{rd} \text{ hour} &= 6000 \times \frac{3}{2} = 9000 \\ 4^{th} \text{ hour} &= 9000 \times \frac{3}{2} = 13500 \\ 5^{th} \text{ hour} &= 13500 \times \frac{3}{2} = 20250 \end{aligned}$$

For the 5 hours, the total money received by the house maid is

$$4,000 + 6,000 + 9,000 + 13,500 + 20,250 = 52,750$$

Qn. 15. The sum of the four scopes $= 42 + 43 + 46 + 49 = 180$

The sum of five scores $= 180 + x$

The possible values of x are:

i) $x < 43$. The possible arrangement of scores in ascending order is

$$X,\ 42,\ 43,\ 46,\ 49 \quad \text{or} \quad 42,\ X,\ 43,\ 46,\ 49$$

such that

$$\begin{aligned} \text{average of the 5 students} &= \text{median of the 5 students} \\ \frac{180 + X}{5} &= 43 \\ 180 + X &= 215 \\ X &= 35 \end{aligned}$$

ii) $x \geq 43$ & $x \leq 46$. The possible arrangement of scores in ascending order is

$$42,\ 43,\ X,\ 46,\ 49$$

such that

$$\begin{aligned} \text{average of the 5 students} &= \text{median of the 5 students} \\ \frac{180 + X}{5} &= X \\ 180 + X &= 5X \\ 4X &= 180 \\ X &= 45 \end{aligned}$$

iii) $x > 46$. The possible arrangement of scores in ascending order is

$$42,\ 43,\ 46,\ X,\ 49 \quad \text{or} \quad 42,\ 43,\ 46,\ 49,\ X$$

such that

$$\begin{aligned} \text{average of the 5 students} &= \text{median of the 5 students} \\ \frac{180 + X}{5} &= 46 \\ 180 + X &= 230 \\ X &= 50 \end{aligned}$$

Therefore, the possible values of X are $X = \{35, 45, 50\}$